Experimental Investigation of Nano-ZrO_2 Reinforced Al Nanocomposites.

B. Tech. Phase II Major Project Report submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology (B. Tech.)

In

MECHANICAL ENGINEERING

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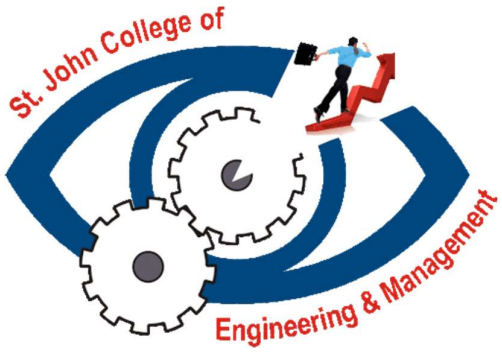
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2026 – 2027

CERTIFICATE

This is to certify that the B.E. project entitled **". "Experimental Investigation of Nano-ZrO_2 Reinforced Al Nanocomposites"** is a Bonafide work of **"Mr. Tanish N Shinde (EU2235005)", "Mr. Raman N Tiwari (EU2235011)", "Mr. Yash N Patil (EU2235007)"** and "**Mr. Dhiraj R Yadav (EU2235015)"** submitted to University of Mumbai in partial fulfilment of the requirement for the award of the degree of **"Bachelor of Technology"** in **"Mechanical Engineering"** during the academic year 2026 – 2027.

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Examiners

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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It expresses my profound gratitude to my Research Guide Dr. Jagdish Pawar
Assistant professor, Department, St. John College of Engineering and Management,
Palghar enabling me to complete the project report…

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Abstract

The aim of this project is to conduct an experimental study on aluminum metal matrix composites reinforced with zirconium oxide (ZrO_2) nanoparticles. The composite specimens are produced by the stir-casting method in which the powder of ZrO_2 is melted into the molten aluminum. In this regard, different quantities of reinforcement, stirring speeds and stirring times are taken into consideration to analyze their influence on the properties of the manufactured specimens. Nine experimental trials are planned and performed using a Taguchi L9 orthogonal array.An L9 orthogonal array is used to plan and perform the nine experimental trials. Hardness values of the prepared specimens are tested in Rockwell hardness testing machine. Experimental readings are taken and analysed and the performances of the specimens are compared with different processing conditions. The purpose of the study is to find out how the parameters selected affect the hardness of the aluminum–ZrO_2 composite and to determine the optimum parameters for the composite fabrication.

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List of Abbreviations

Al	**Aluminum alloy**
ZrO_2	**Zirconium oxide**
MMC	**Metal Matrix Composite**
MMNC	**Metal Matrix Nanocomposite**
DOE	**Design of Experiments**
L9	**Taguchi L9 Orthogonal Array**
OA	**Orthogonal Array**
SEM	**Scanning Electron Microscopy**
EDS	**Energy Dispersive Spectroscopy**
UTS	**Ultimate Tensile Strength**
HV	**Vickers Hardness**
XRD	**X-Ray Diffraction**
rpm	**Revolutions per minute**
wt.%	**Weight percent**
vol.%	**Volume percent**
MS	**Mild Steel**
SS	**Stainless Steel**
T6	**Solution treatment/aging designation where applicable**

CHAPTER 1
INTRODUCTION

Aluminum metal matrix composites (AMMC's) are not only lightweight and strong but also have been a subject of extensive research in the engineering community because they may also provide improved mechanical properties. While Al is a strong alloy that provides good strength-to-weight ratios in many applications; enhancing its hardness and wear resistance could make it even better. ZrO_2 is a ceramic material which has been known to have superior hardness and thermal stability property. So, it could be a promising reinforcement in Al alloys. The aim of this study was to find out how adding nano-ZrO_2 powders to Al matrix affects the hardness of the composite.

In order to prepare the reinforced composites, the method of stir casting is used with ZrO_2 as reinforcement to the molten Al. In particular, the mixture of the reinforcing materials with molten metal can be done by means of motor-driven stirrer that helps to carry out the mixing process. Different reinforcement ratios, stirring speeds and stirringer times are selected as the parameters to study the effect of these parameters on the properties of the specimens. A Taguchi L9 orthogonal array is used to perform nine experiments, to design the experimental process.

The hardness of produced samples is determined by using Rockwell hardness test and the data acquired are compared further. The aim of this work is to gain an understanding of the feasibility of producing.

The aluminum alloy was prepared by stir casting and the effect of nano-ZrO_2 on the characteristics of the aluminum alloy was studied in Al/ZrO_2 nanocomposites.

1.1 Motivation

The growth in use of lightweight materials in auto, aerospace, transport and general engineering fields has seen a demand for materials which perform well without added component weight. Due to their low density and good engineering properties, the Aluminum alloys are of great interest in these areas. But also our conventional aluminum alloys fall short in terms of hardness, wear resistance and other properties which see heavy use. Aluminum's high strength is what makes it a great play in structural and engineering applications. That said improvement in certain properties may make the material a better fit for applications which include friction, surface contact and mechanical loading.

Also, ceramic reinforcement in an aluminum matrix is a “put forth” method to observe these improvements.

In the present work we chose ZrO_2 which is a ceramic material for its hardness, wear resistance and thermal stability. We used nano sized form which although small in volume has a large surface area which in turn makes particle dispersion in the composite preparation process a key issue.

Stir casting process was chosen as a process that we found to be a comparatively easy technique for producing particulate metal matrix composites and is possible to be undertaken in the lab scale.

This project was also selected for our efforts, as we wanted to explore a real-world application for the manufacture of Al composite reinforced with nano ZrO2 and to investigate how varying the amount of reinforcement affects the overall properties of the composite.

1.2 Problem Statement

While the low density and high strength make aluminum alloys extremely useful, there are instances where a higher level of hardenability, wear resistance and other mechanical improvements are required, and in such cases, their performance falls short.

It is also observed that uniform production of aluminum based nanocomposite is a challenge. At nano scale the ZrO2 particles have a tendancy to aggregate due to their small size and large surface area.

Also we have the issue of finding out proper process for the incorporation of nano ZrO2 into Al using stir casting and to look at the effect of various reinforcement levels.

In this work we directly address the issue of ZrO2 volume fraction in Al, we investigate the effects of such out experimentally.

1.3 Objectives

1. The major objectives of this project are to.

2. To research the mechanical properties of Al alloy and nano-ZrO2 strengthening material.

3. To prepare Al/ZrO_2 nanocomposite using stir casting technique.

4. To prepare composite specimens that we looked at which had 0.8gm, 1.1gm and 1.6gm by volume of ZrO2 in them for our experiments.

5. Another thing that we set up and ran was a laboratory scale stir cast facility, this included an electric resistance furnace, graphite crucible, mechanical stirrer, speed control feature and anything else that we required for the preparation of the composites.

6. Also we did a study to see what effect ZrO2 addition has on Al's properties which we looked at via the right tests we did on the made specimens.

7. To observe the effect of processing and reinforcement parameters on quality and character of the composite produced.

8. Also we will use the Taguchi L9 experimental design for a systematic approach to the set up and analysis of the experiments.

9. Also we will look at the results obtained from the different reinforcement levels and note which changes are brought about.

10. To test the suitability of using stir casting technique for preparing Al composites at laboratory scale containing nano-ZrO_2.

1.4 Scope

In the present project, the experimental development and investigation of nano-ZrO_2 reinforced Al aluminum matrix nanocomposites produced by the stir casting technique is limited. The project involves:

- Welding and processing of Al matrix material. Use of nano-ZrO_2 powder as ceramic reinforcement.

- The production of composites using stir casting technique. More investigation of volume fractions of 0.8gm, 1.1gm and 1.6gm ZrO_2.

- The use of a laboratory scale electric resistance furnace and graphite crucible.

- Mechanical stirring of molten composite.
- Production of casts using suitable mould.
- Application of Taguchi L9 design for experimental design. In addition, there has been experimental testing of the
- To prepare specimens using the proper material property test.
- Comparison of the observations obtained for different reinforcement conditions.
- Investigation of the possibility to use selected casting method for the production of Al/ZrO_2 nanocomposites.

This work places more importance on the preparation of materials in the laboratory and experimental study.

Chapter 2
REVIEW OF LITERATURE

Aluminum-based metal matrix composites have been given much focus as it is possible to tailor the properties of aluminum alloys by adding appropriate reinforcement materials. The ceramic material Zirconium oxide (ZrO_2) is considered for a reinforcement due to its high hardness, wear resistance, thermal stability and other suitable mechanical properties. The following studies were reviewed to understand the behaviour of Al/ZrO_2 composites effect of reinforcement content and importance of processing parameters.

2.1 Sivakumar et al. (2023)

This study aims to investigate the mechanical properties of Al aluminum alloy reinforced with ZrO_2 through stir casting. Four compositions were thought of, namely the pure Al and the Al reinforced with 5%, 10% and 15% ZrO_2. The researchers concentrated on testing tensile strength, impact strength and hardness. The study presents the explanation of how ceramic reinforcement can be introduced in the molten aluminum matrix by mechanical stirring. This research is related to the present project because it has the same matrix material, reinforcement and manufacturing method as the present project.

2.2Sharma, Mallick and Gupta (2023)

Sharma, Mallick and Gupta studied the mechanical and tribological performance of Al alloy with nano-ZrO_2 reinforcement. The nanocomposites were produced by the powder metallurgy technique, which involves ball milling, compaction, sintering and hot extrusion. Different ZrO_2 reinforcement levels of 0, 1.25, 2.53, 3.84, and 5.19 vol.% were studied. The results indicated that tensile and yield strength of the material had significantly improved at 2.53 vol.% ZrO_2 whereas wear resistance had significantly improved until 5.19 vol.%. The findings of this research will be useful to understand the effect of ZrO_2 content on the properties of Al composite.

2.3 Annapoorna et al. (2024/2025)

Annapoorna and co-authors investigated the optimized parameters of stir casting of Al6061 aluminum alloy reinforced with nanoalloys$_3$ and nano-ZrO_2 particles. The hybrid metal matrix composites were prepared by the ultrasonic assisted stir casting method. The parameters of reinforcement composition, stirring temperature, stirring speed and stirring time were investigated using a Taguchi orthogonal array of L16. The optimization of ultimate tensile strength was done by signal to noise ratio analysis, regression modelling and software Minitab. The study emphasizes the need to choose appropriate processing parameters to enhance mechanical properties of aluminum based composites.

2.4 Jagdish Pawar (2022)

Pawar and Tungikar studied process parameters optimization for stability of an alumina–distilled water-based nanofluid. The study analyzed factors including: concentration of the nanoparticles, particle size, sonication time, initial stirring time, and concentration of the surfactant. Experiments were designed using Taguchi L18 orthogonal array and the number of trials were minimized. Signal to noise ratio analysis, ANOVA and regression modelling were used to determine the influential parameters and to predict the stability of nanofluid. In this study, the researchers used a nanofluid instead of Al–ZrO_2 composites, but the study can serve as a guideline for using the Taguchi method in optimization of experimental parameters.

CHAPTER 3
REQUIREMENT ANALYSIS

The materials, processing equipment, supporting components and analysis methods are needed for the development of nano-ZrO_2 reinforced Al nanocomposites. The requirement analysis is used to identify the hardware and software requirements, functional and non-functional requirements for the successful implementation of the experimental investigation. The current project deals with the preparation of Al/ZrO_2 nanocomposites by stir casting method.

The main components of the major requirements are an electric resistance furnace to melt the Al alloy, a graphite crucible to hold the molten metal, a mechanical stirring arrangement to incorporate the nano-ZrO_2 reinforcement and a mould to produce the cast specimens.

Besides, facilities for measuring and data analysis are needed to obtain the amount of materials, record experimental parameters and analyse test results.

3.1 Hardware Requirements

The hardware needs are those of the equipment, materials and components that are needed to prepare the Al/ZrO_2 nanocomposite. Hardware / Material Used

1. Electric Resistance Furnace: This is for the purpose of heating and melting the Al alloy before adding reinforcement.



Fig 3.1.1 Electrical Furnace

2. Graphite Crucible:
Used to contain the Al alloy during heating and melting.



Fig3.1.2 Graphite Crucible**.**

3. Al Alloy:
Used as the matrix material for the composite.



Fig 3.1.3 Aluminum Bar

4. Nano-ZrO_2 Powder:
Used as the ceramic reinforcement in the Al matrix.



Fig 3.1.4 Nano Zr02 Powder

5. Mechanical Stirrer

Used as the ceramic reinforcement in the Al matrix.



Fig 3.1.5 Mechanical Stirrer

6. **Mould:**
   Used for pouring and solidification of the molten composite to obtain cast specimens.

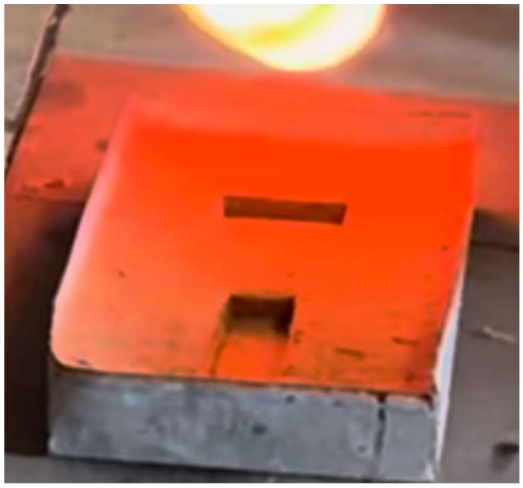


Fig 3.1.6 Mould

7. **Digital Weighing Balance:**
   Used for measuring the required quantities of Al and ZrO_2.



Fig 3.1.7 Weighing Balance

8. Personal Protective Equipment:
Required for safe handling of hot metal and casting operations.



Fig 3.1.8 Safety Equipments

9. Brinell Hardness Tester:
Required for Checking Hardness Of the new alloy.



Fig 3.1.8 Safety Equipments

Description

Al alloy is melted mainly by the use of the electric resistance furnace. The alloy is poured in the graphite crucible which is capable of holding the molten alloy during the melting and stirring processes.

To introduce and distribute the nano-ZrO_2 particles in the molten Al, a mechanical stirring system is needed.

The stirring system consists of a motor, speed controller, bearing, stainless-steel shaft and graphite impeller.

The motor gives the stirring motion and the required stirring speed is maintained by the speed controller.

Therefore, the mould is an important component of the casting system since it sets the basic shape and size of the specimen being cast. Also, the mould and cast specimens should be measured with the appropriate tools to ensure the size of the moulds and castings are correct.

3.2 Software Requirements

There is no need to use special software in the present project in order to control the casting device. However software based tools can be utilized in design calculations, dimensional modelling, experimental planning and analysis of experimental data..

Table 3.2 Software Requirements

Sr. No.	Software / Tool	Application
1	Microsoft Excel / Spreadsheet Software	Calculations, preparation of experimental tables and analysis of recorded data.
2	CAD Software	Preparation and modification of mould/component designs where required.
3	Minitab Software	Planning and analysis of the selected Taguchi experimental design, if available.

3.3 Functional Requirements

Functional requirements outline the operations that the proposed experimental setup and project methodology is expected to be able to perform.

Material Measurement
The system should be able to accurately measure the required amounts of Al alloy and nano-ZrO_2 powder, depending on the composition of the experiment conducted.

Melting of Al
The furnace should be able to melt the Al alloy to the desired temperature to

allow the reinforcement to be added to the molten alloy.

Reinforcement Addition
The setup shall enable the desired amount of nano-ZrO_2 powder to be added to the melted Al under the desired experimental condition.

Mechanical Stirring
The stirring mechanism should be designed to generate rotational movements to the graphite impeller so that nano-ZrO_2 particles are mixed with the molten Al.

Stirring Speed Control
The speed-control arrangement shall enable the stirring speed selected to be operated at the required speed as planned in the experiment.

Composite Casting
The molten Al/ZrO_2 composite after the required melting and stirring shall be poured into the mould prepared.The molten Al/ZrO_2 composite after the required melting and stirring shall be poured into the mould prepared for solidification.

Specimen Preparation
The casting process should yield specimens of suitable dimensions and quantity of material to perform the experimental tests.

Experimental Planning
The project will enable the selected process parameters and reinforcement levels to be arranged following the Taguchi L9 experimental design.

Data Recording
All the experimental conditions and the observations collected during each experimental run shall be systematically recorded to be analysed.

Result Comparison
The experimental data will be compared to each other for the various reinforcement conditions of ZrO_2 and the various process parameters to find their effects on the chosen properties.

3.4 Non-Functional Requirements

Non-functional requirements specify the quality, safety, reliability and Operational conditions which should be kept during the experiment.

Safety
Care shall be taken in operating the experimental setup as it contains hot equipment as well as high temperature molten aluminum. The appropriate protective equipment should be worn when operating the furnace, stirring and casting**.**

Repeatability
The experimental procedure should be done in a uniform manner so that the results of the different runs of the experiment can be meaningfully compared.

Dimensional Accuracy
The mould and specimens must be as close to set sizes as possible. Care must be taken in dimensional control when specimens are then prepared for mechanical or other material tests. Process

Control
For each experimental run the important processing conditions such as quantity, stirring speed and stirring time of the reinforcement, and other experimental parameters should be controlled and recorded**.**

Material Uniformity
The processing method should try to achieve a fairly even dispersion of the nano-ZrO_2 in the Al matrix. Good stirring, therefore, is an important requirement of the experimental procedure**.**

Data Accuracy
The amount of materials, experimental parameters and test observations should be noted carefully, so as to reduce the calculation and observation errors**.**

Cost Effectiveness
The experimental setup, wherever possible, should be composed of commercially available components and laboratory-scale equipment in order to allow the production of the nanocomposite within the project budget.

CHAPTER 4
DESIGN

4.1 Requirement Elicitation

The process of determining the materials, equipment, experimental parameters and testing requirements required to successfully implement the project is referred to as requirement elicitation.

4.1.1 Use Case Diagram

The use case diagram shows the main activities carried out by student/operator in the experimental investigation. It explains interaction between operator and experimental system.

Main functions:

- Designing the experiments by the Taguchi L9 method.
- Measuring the weights of Al alloy and ZrO_2 reinforcement.
- Melting the aluminum alloy in the furnace.
- Adding reinforcement and stirring the molten material.
- The pouring of composite melt into the mould. Collection and identification of the specimens.
- Hardness measurement by HRB scale.
- Collecting data from experiments and analysing them.

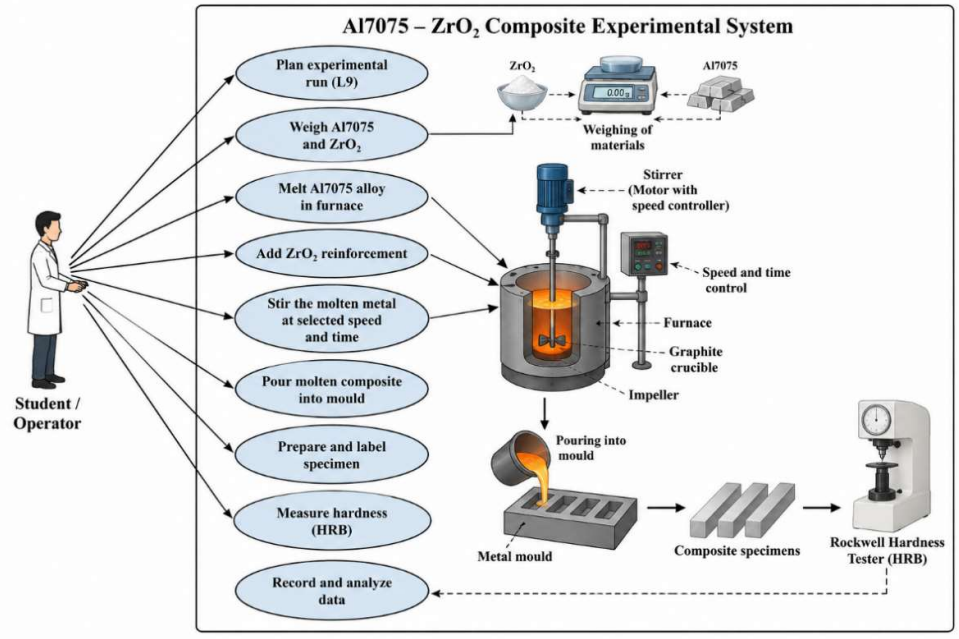


Figure 4.1.1: Use Case Diagram of the Al–ZrO_2 Composite Investigation

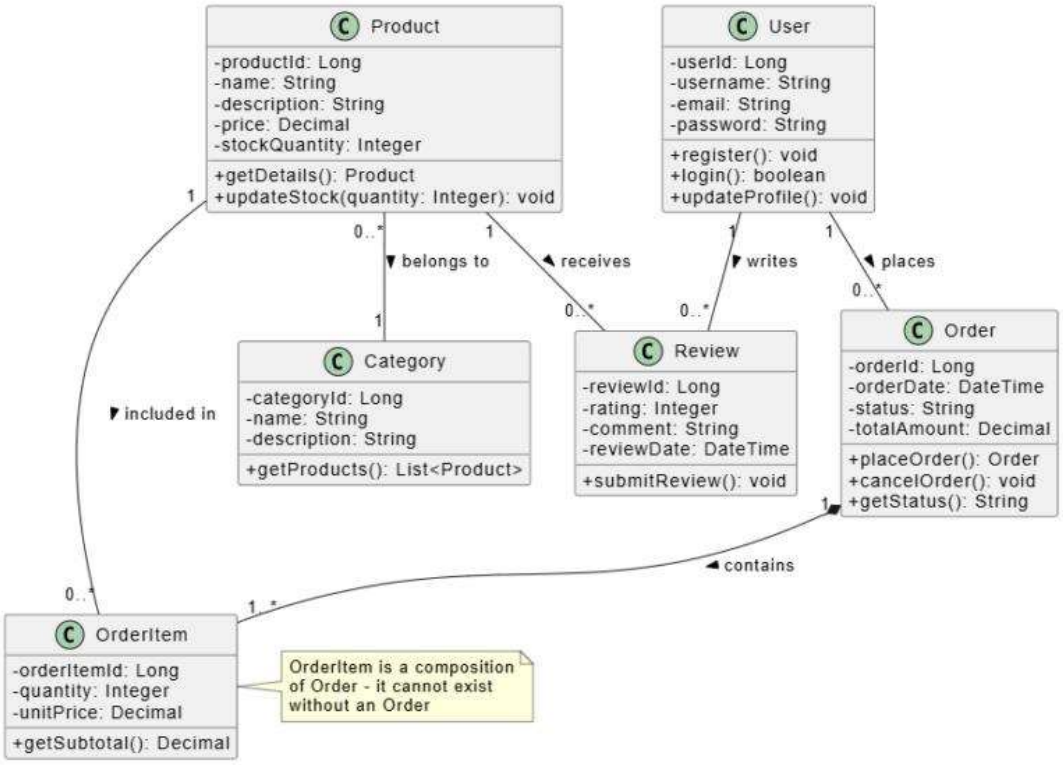


Fig 4.1.2 Class Diagram.

4.1.2 Class Diagram

1. Material Batch: Records the amount of Al alloy and ZrO_2 reinforcement used in each experimental test.
2. Stir Casting Setup: This is a representation of the furnace, graphite crucible, motor, speed controller, shaft and impeller used in the stir casting process.
3. Specimen: Records the sample identification, mould and casting information.
4. Hardness Test: Stores the testing scale, hardness readings and test information.
5. DOE Analysis: Stores the Taguchi L9 factor levels, experimental run details and hardness response.

4.1.3 Activity Diagram

The activity diagram shows the sequence of activities of manufacturing and testing the Al–ZrO_2 composite specimens.

The steps towards the process include choosing a particular experimental run, weighing the necessary materials. Preparation of furnace and crucible, melting of Al alloy.

A given amount of ZrO_2 reinforcement is added to the melt and then stirred at the desired speed and time. The melted composite is then stirred and poured into the mould and cooled down. The sample is then taken off after safe cooling and labelled and prepared for hardness testing. Finally, the hardness reading is noted down corresponding to each experimental run and the process is repeated for each of the 9 runs.

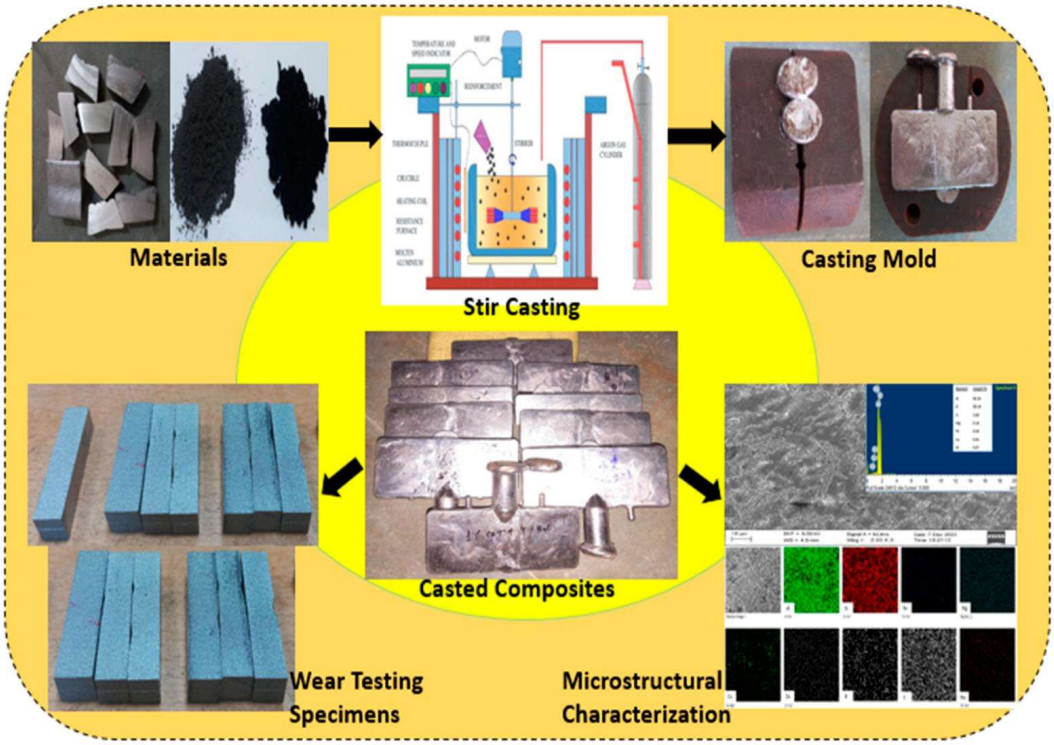
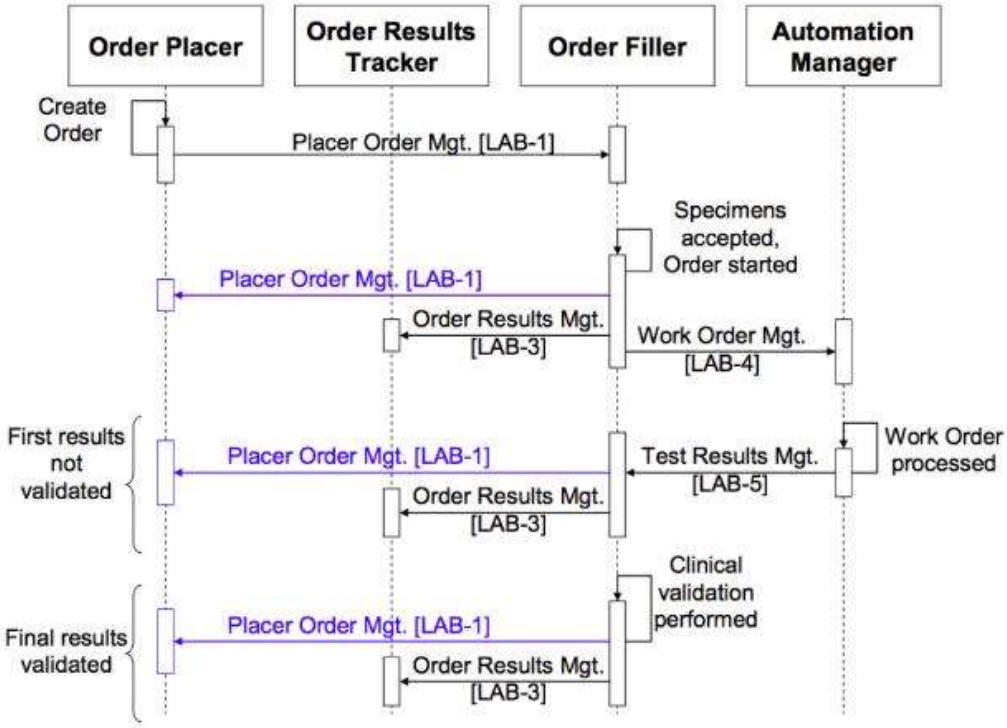


Fig 4.1.3 Activity Diagram.

4.1.4 Sequence Diagram

The sequence diagram shows the flow of interactions between the operator and the main parts of the experiment during one experimental run. At first, the operator weighs the alloy and reinforcement, and pre-heats the furnace and crucible. The aluminum alloy is melted and the stirrer is activated under the experimental condition selected. Next the molten composite is poured into the mould. Once solidified and prepared, the operator uses the hardness tester to determine the HRB value. The reading is noted on the appropriate run number for later analysis.



Fig

4.1.4 Sequence Diagram.

4.2 Data Flow Diagrams

Data Flow Diagrams (DFDs) are used to depict the flow of data into, through and out of the experimental investigation. The input information in this project consists of quantities of materials, the level of reinforcement, the speed of stirring, the stirring time and test conditions.

4.2.1 Level 0 DFD

Level 0 DFD is a view of the entire project as a single process. The amount of materials, the experimental factor settings and the test conditions are fed to the Al-ZrO2 composite investigation system. The outputs are prepared specimens, recorded HRB value and results of the experimental analysis..

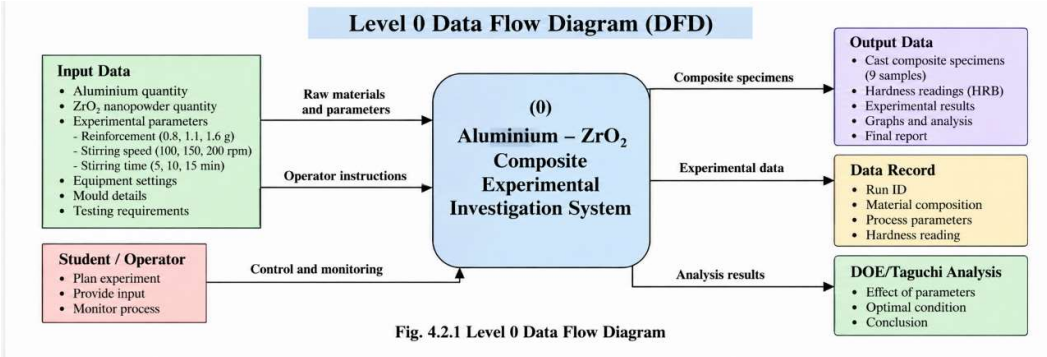


Fig. 4.2.1 Level 0 Data Flow Diagram

4.2.2 Level 1 DFD

The Level 1 DFD breaks the main process down into steps. The amount of material and experimental conditions are noted. The Al alloy is next melted, the reinforcement is added and the melt is agitated. The composite is cast in a mould and the specimen is ready for hardness testing. The measured hardness data is checked and put in the experimental data table. Lastly, analysis of the results is carried out with Taguchi L9 method and suitable plots are produced for comparison of the experimental runs.

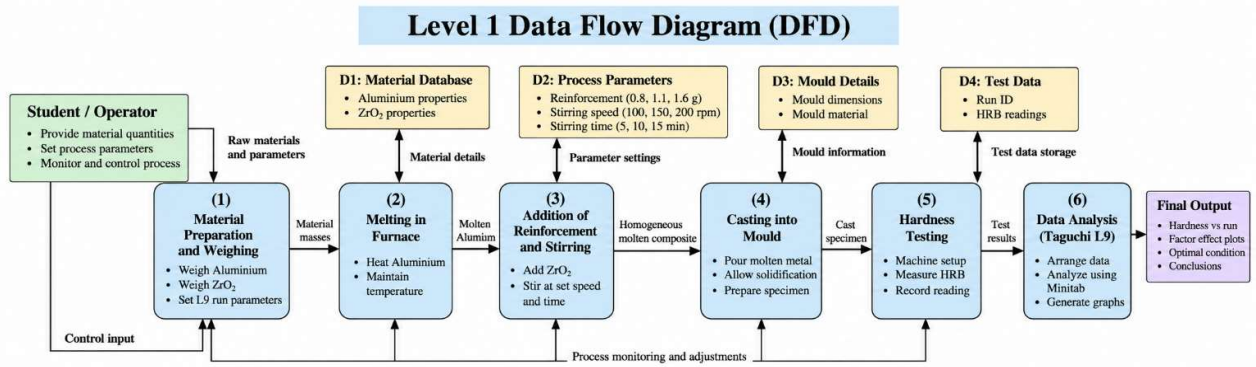


Fig. 4.2.2 Level 1 Data Flow Diagram

4.3 Timeline / Gantt Chart

Gantt chart displays the main activities that were performed during the project, from the literature review to the preparation of the report. It involves the procurement of materials and equipment, development of the mould, casting of nine specimens, hardness testing, and analysing the test results. The following Schedule is a planning chart, which is indicative only. Before handing in, substitute the week number with actual dates as found in your project diary.

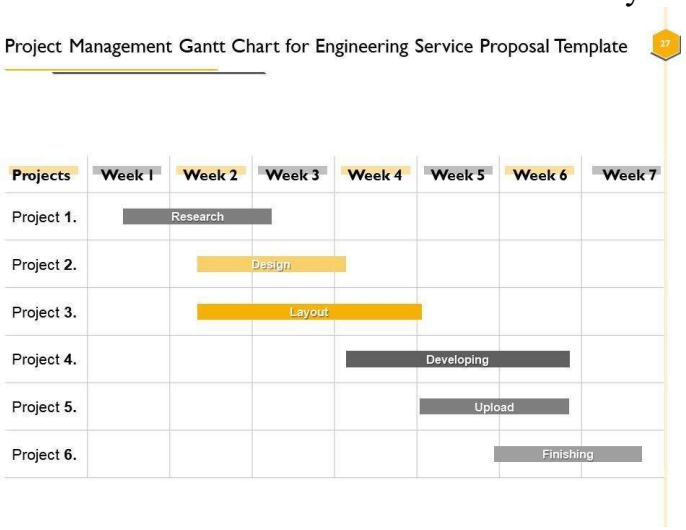


Fig 4.3 Timeline Graph Diagram.

4.4 Work Breakdown Structure (WBS) Chart

The Work Breakdown Structure breaks down a project into smaller tasks, which helps to plan, execute and monitor the project systematically. The project is organised into the following key work packages:

1. Planning: Objectives are decided and literature is reviewed.
2. Material procurement: Procurement and weighing of Al alloy and ZrO_2 nano powder.
3. Preparation for set up: Arranging the furnace, crucible, motor, shaft, and impeller.
4. Experimental work: Melting, reinforcement addition, stirring, casting and preparation of specimen.
5. Testing: Hardness test and note taking of HRB readings.
6. Data Analysis using Taguchi L9 analysis and preparing graphs.

Documentation: Discussion, conclusion, references, plagiarism/similarity checking

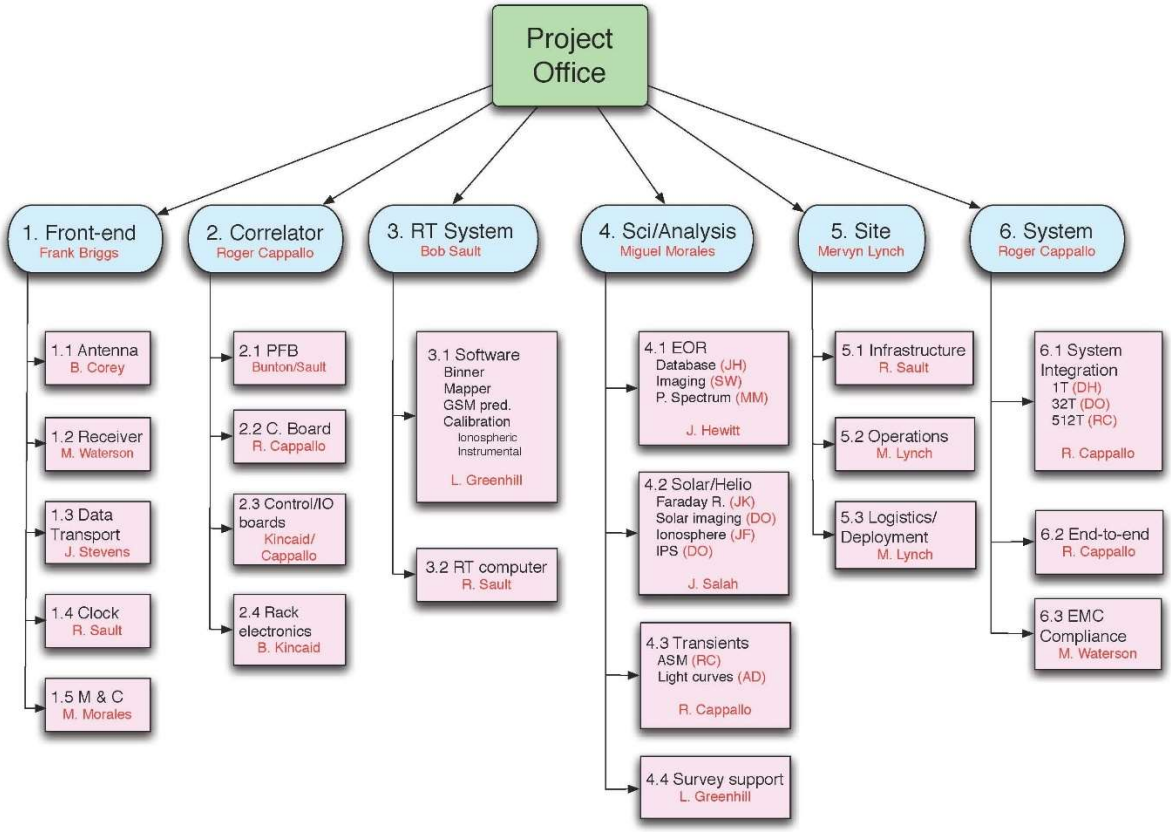


Fig 4.4 Work Break down Diagram.

Chapter 5
Report on Present Investigation

5.1 Proposed System

his proposed system involves fabrication and experimental studies of Al aluminum alloy reinforced with ZrO_2 nanoparticles through stir casting process.

The primary aim is to investigate the influence of the quantity of reinforcement, speed and period of stirring on the hardness of the prepared composite pieces.

The experimental setup consists of an electric resistance furnace, graphite crucible, a motor driven stirring mechanism, stainless-steel shaft and graphite impeller and mould. Measured amount of ZrO_2 reinforcement is added to the molten metal in the furnace.

The mixture is stirred based on the desired experimental conditions and poured into a mould. The specimens are prepared and examined after solidification with a Rockwell hardness tester (HRB) scale.

The recorded readings are arranged in the order of Taguchi L9 experimental design and analysed to understand the effect of the selected process parameters.

The proposed system is systematic in material preparation, melting, adding reinforcement, stirring, casting, hardness testing, recording and analysis.



Fig 5.1 Proposed System.

The goals of the suggested system

1. To produce Al-ZrO_2 composite samples by stir casting method. To determine how the amount of reinforcement affects the hardness
2. To investigate the effect of stirring speed and stirring time.
3. To determine the hardness of the specimens prepared and record it.
4. To carry out an analysis of the experimental observations based on Taguchi L9 approach.
5. To determine suitable process parameter settings from the measured results**.**

5.2 Methodology

The fabrication and experimental studies of Al–ZrO_2 nanocomposites is done by the stir-casting method. Al aluminum alloy and ZrO_2 nano powder were first chosen and weighed based on the experimental design. Al alloy was melted using graphite crucible in an electric resistance furnace. The amount of ZrO_2 reinforcement was measured and subsequently added to the molten alloy, which was stirred with the help of a motor-driven stirring mechanism for the desired time at the aforementioned speeds.

The resulting molten composite material was then poured into the mould and left to cool. The fabricated specimens were then taken out and prepared and labelled with their experimental run numbers. The HRB scale of the Rockwell hardness tester was used to measure the hardness of each specimen. Lastly, the hardness values obtained were measured and then examined using the Taguchi L9 experimental design with the aim of investigating the effect of the various amounts of reinforcement, stirring speed and stirring time on the hardness of the fabricated composites.

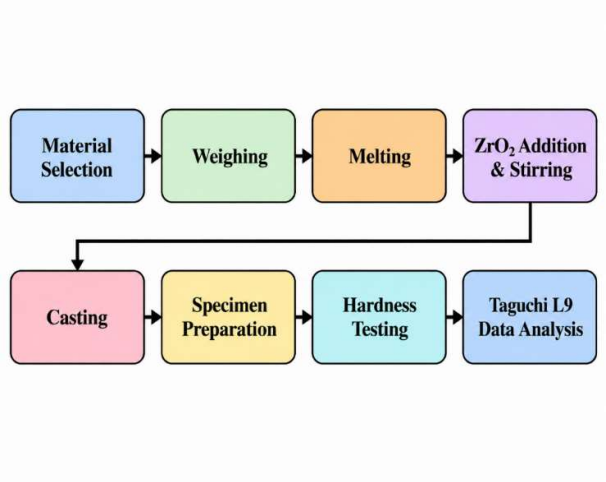


Fig5.2 Methodology Diagram.

5.3 Calculations

Calculation of Al Required for One Sample

Given:

Length	=	75	mm
Width	=	30	mm
Depth	=	15	mm

Density of Al = 2.81 g/cm³

Step 1: Calculate Volume
Volume = Length × Width × Depth
Volume = 75 × 30 × 15
Volume = 33,750 mm³
Since 1 cm³ = 1000 mm³,
Volume = 33,750 / 1000
Volume = 33.75 cm³

Step 2: Calculate Mass of Al
Mass = Volume × Density
Mass = 33.75 × 2.81
Mass = 94.84 g

Final Result:

The amount of Al needed for one sample is about 94.84 g. Theoretically, the amount of Al to be used for one sample was calculated to be approximately **94.84 g.**

To allow for some loss of Al during melting, stirring, pouring, oxidation, handling, and machining, however, an additional 50-60 g of Al was added.

Hence, **145–155 g** Al was taken for the preparation of each sample.

Chapter 6

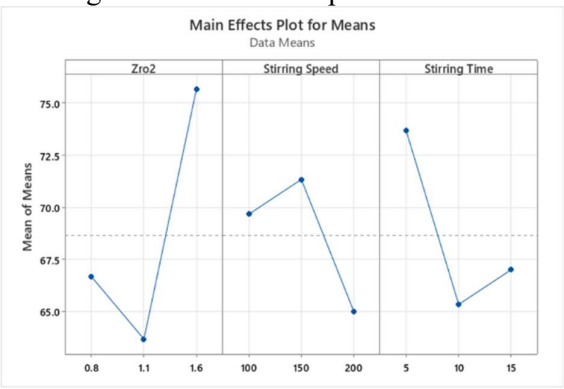
Result and Discussion.

Zro2 (wt%)	Stirring speed (rpm)	Stirring time (sec)	Hardness (HRB)
0	0	0	40
0.8	100	5	74
0.8	150	10	66
0.8	200	15	60
1.1	100	15	60
1.1	150	10	66
1.1	200	5	65
1.6	100	10	75
1.6	150	5	82
1.6	200	15	70

Table 6.1 Result Table

The main effects plot illustrates the effect of different amounts of reinforcement (ZrO_2), stirring speed and stirring time on the mean hardness of the Al–ZrO_2 composite. The above results show that the quantity of ZrO_2 (1.6 g) gave the highest mean hardness; stirring speed (150 rpm) also gave the highest mean hardness and stirring time (5 minutes) gave the highest mean hardness. The quantity of ZrO_2 has the greatest mean hardness deviation, indicating that it has the greatest effect on the experimental response. The values obtained from the plotted means seem to be satisfactory for attaining higher hardness for the tested conditions.

Fig 6.1.1 Main Effect plot for means



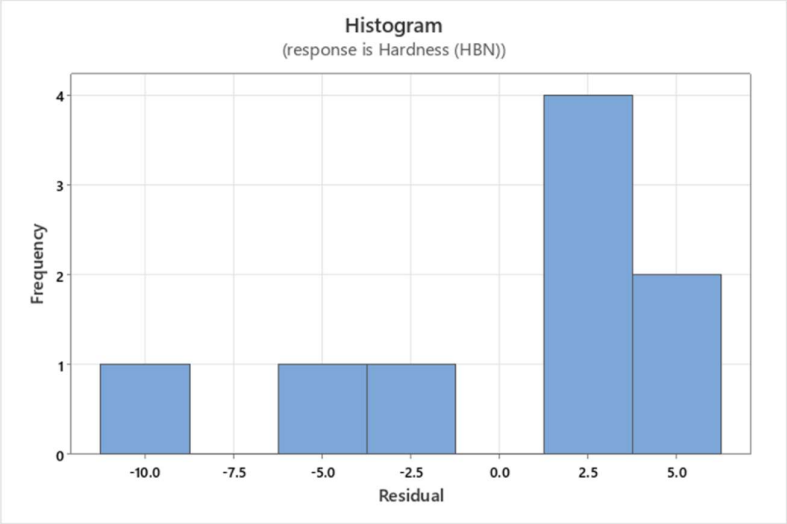


Fig 6.1.2 Histogram Hardness

Residual Histogram is the distribution of the errors or differences between the predicted and observed hardness values. The majority of the residuals are on the positive side with a few negative residuals too. This plot helps assess the residual distribution and the suitability of the fitted model for analyzing the hardness data.

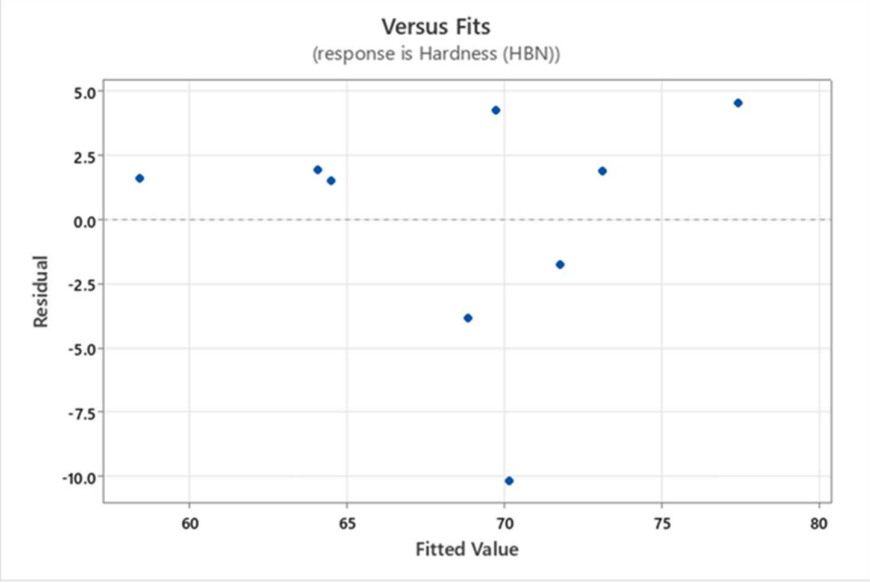
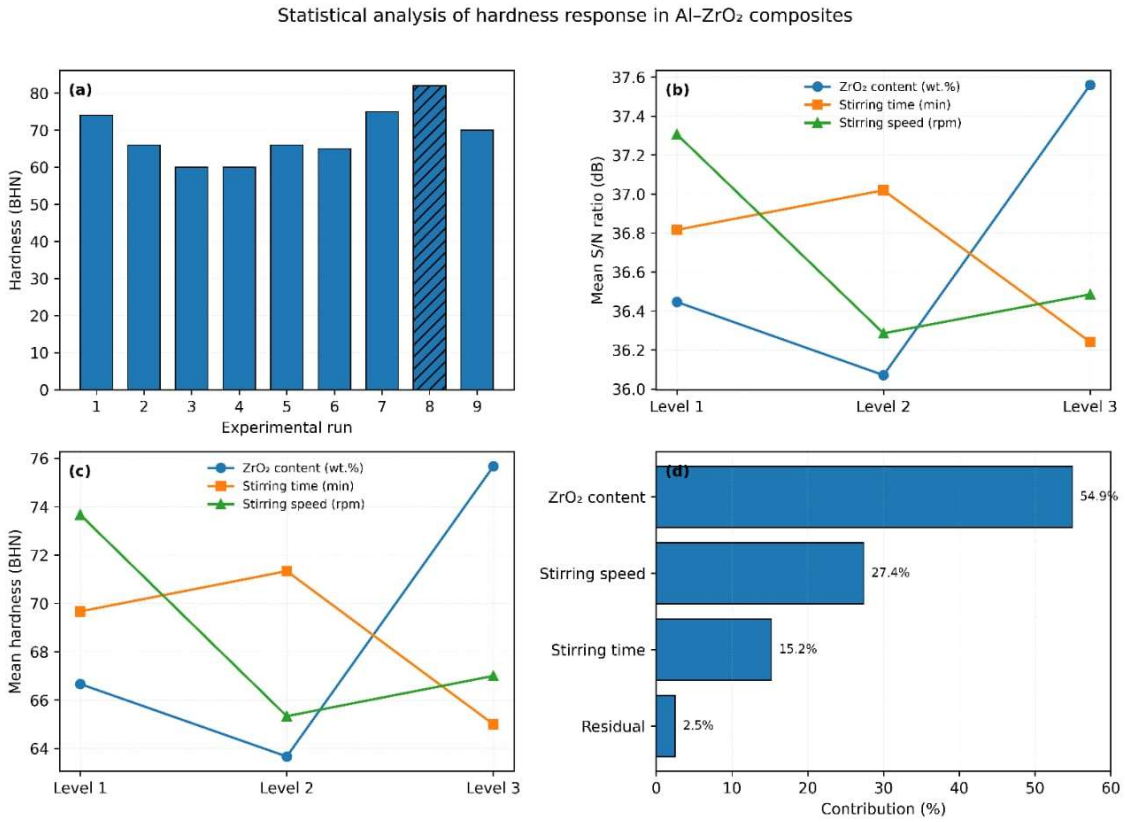


Fig 6.1.3 Residual V/S Fitted Value Hardness

The residuals versus fitted values plot shows the difference between the observed and predicted hardness values. The residuals are spread around the zero line, although there are some relatively large deviations. This means there's some uncertainty in the model predictions and should be seen with great



caution.

Fig 6.1.4 Different Graphs for Hardness Values Comparison.

The Minitab analysis of the nine Al–ZrO_2 composite samples' hardness results was done using the Taguchi L9 design. The experimental hardness results were 60 HRB to 82 HRB, the highest hardness of 82 HRB was obtained for Run 8 (1.6 g ZrO_2, 150 rpm stirring speed and 5 s stirring time).

Based on the Taguchi analysis for the “Larger is better” S/N ratio, the influence of the ZrO_2 content is greatest for the hardness, followed by the stirring time and stirring speed.

Importance of Taguchi Method

The optimization of the experimental parameters for preparation of nanocomposites like ZrO_2 reinforced aluminum prepared by stir casting process is important where Taguchi method is applied. In this project the Taguchi L9 orthogonal array is used to study the influence of the selected process parameters at three levels. This design can help to decrease the number of experiments in comparison to a full factorial design, which consequently saves experimental time, material and cost. Signal-to-Noise (S/N) ratio may be used to detect combinations of parameters that will enhance the desired response, such as increase hardness. In addition, analysis of variance (ANOVA) can be used to find out the relative contribution of individual process parameters to hardness variation. Therefore, the Taguchi method is a systematic, economical and efficient method to be used in experimental planning, process optimization as well as to improve the quality and uniformity of the developed nanocomposite. The results can be used to select appropriate stir casting parameters to get desired mechanical properties.

In this project, the Taguchi method has been applied to minimize the number of experiments and to achieve the economical and efficient investigation of the project. Three process factors with three levels were taken into consideration. A full factorial design would involve $3^3 = 27$ experiments to study the various combinations of factors selected. However, with the Taguchi L9 orthogonal array, the experimental design was decreased to just nine experimental runs. This is 18 experiments or about 66.67% less experimental runs. Therefore, the method saved precious experimental time, raw materials, energy and total testing costs. It also gave a systematic study for the effect of process parameters selected on the hardness of the prepared aluminum based nanocomposite. Thus, the Taguchi method was useful to plan experiments efficiently, while also providing valuable information on process parameter effects with fewer experimental trials.

Chapter 7
Conclusion And Future Scope.

7.1 Conclusion

The effect of reinforcement amount, stirring speed and stirring time on the hardness of the composite was successfully studied in the 1st person to investigate the Al–ZrO_2 composite specimens with the stir-casting process.

1. The experimental design used was a Taguchi L9 experimental design where three process parameters were varied at three levels (total nine experimental runs).

2. Based on the assumed order of the experiments, the maximum hardness recorded was 82 HRB for Run 8 (1.6 g ZrO_2 reinforcement, 150 rpm stirring speed, and 5 min stirring time)

3. The lowest recorded hardness was HRB = 60; seen in Runs 3 and 4. This variation suggests the processing conditions chosen can affect the hardness of the specimens produced.

4. The calculated mean hardness of nine specimens was around 68.67 HRB in order to have overall experimental result.

5. Experimental investigations facilitated the assimilation of practical knowledge regarding furnace operation, molten-metal handling, reinforcement addition, mechanical stirring, mould casting and specimen preparation.

6. Rockwell B hardness testing gave measurable results to compare the specimens produced under the different experimental conditions. From the main effects plot, it can be seen that the factor levels of 1.5 g reinforcement, 150 rpm stirring speed and 5 min stirring time were related to the maximum mean hardness in the plotted data.

7. These settings need to be further statistically tested before it can be stated that these are the global optimal settings.

8. The Taguchi L9 method was used to plan experiments and compare the effect of various process parameters using only nine experimental runs as

demonstrated in the project.

9. Overall, the project gave the researcher great experimental experience in the fabrication of aluminum metal-matrix composites, hardness measurements and analysis of their data, and gave a foundation for further study of Al–ZrO_2 composites.

7.2 Future Scope.

- ✓ Optimization using Design of Experiments (DOE) and Response Surface Methodology (RSM).
- ✓ Corrosion and fatigue testing.
- ✓ Production of aluminum nanocomposites on an industrial scale.
- ✓ Use in the aerospace, automotive and defence industry.

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